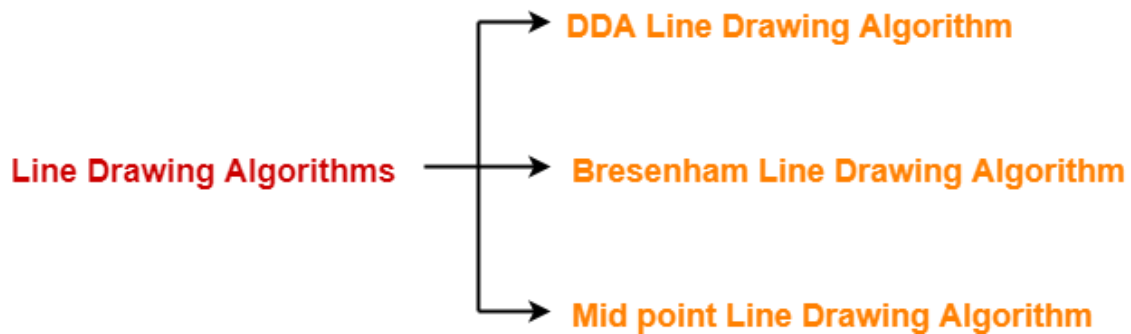


Line Drawing Algorithms

In computer graphics, popular algorithms used to generate lines are-



1. Digital Differential Analyzer (DDA) Line Drawing Algorithm
2. Bresenham Line Drawing Algorithm
3. Mid Point Line Drawing Algorithm

DDA Algorithm is the simplest line drawing algorithm.

Given the starting and ending coordinates of a line,
DDA Algorithm attempts to generate the points between the starting and ending coordinates.

Procedure-

Given-

- Starting coordinates = (X_0, Y_0)
- Ending coordinates = (X_n, Y_n)

The points generation using DDA Algorithm involves the following steps-

Step-01:

Calculate ΔX , ΔY and M from the given input.

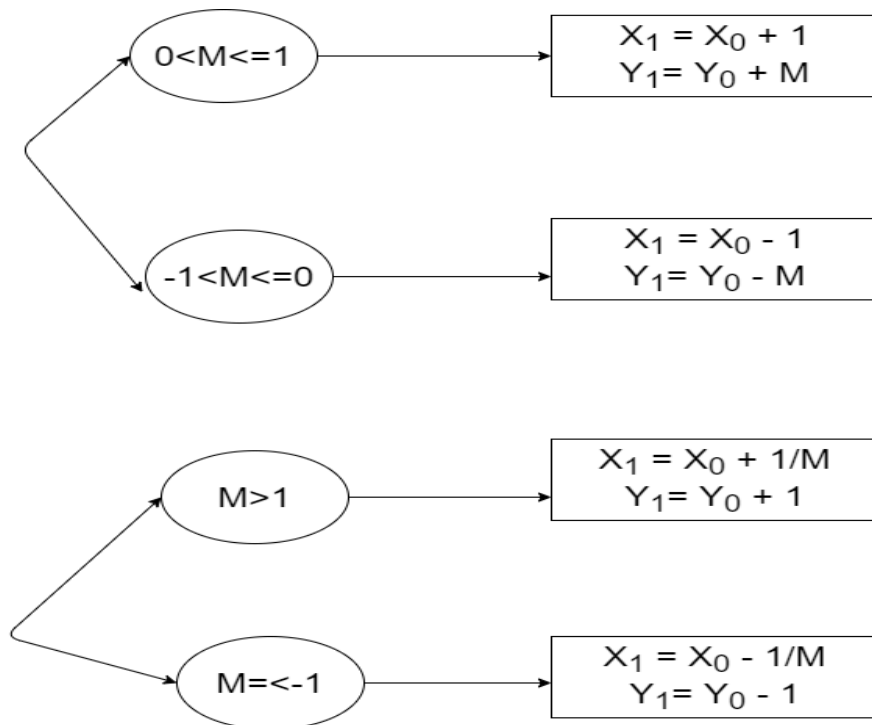
These parameters are calculated as-

- $\Delta X = X_n - X_0$
- $\Delta Y = Y_n - Y_0$
- $M = \Delta Y / \Delta X$

Step-02:

Suppose the current point is (X_0, Y_0) and the next point is (X_1, Y_1) .

Find the next point by following the below four cases-



Step-03:

Keep repeating Step-02 until the end point is reached or the number of generated new points (including the starting and ending points) equals to the steps count

Problem-01:

Calculate the points between the starting point (5, 6) and ending point (8, 12).

Solution-

Given-

- Starting coordinates = $(X_0, Y_0) = (5, 6)$
- Ending coordinates = $(X_n, Y_n) = (8, 12)$

Step-01:

Calculate ΔX , ΔY and M from the given input.

- $\Delta X = X_n - X_0 = 8 - 5 = 3$
- $\Delta Y = Y_n - Y_0 = 12 - 6 = 6$
- $M = \Delta Y / \Delta X = 6 / 3 = 2$

Step-02:

As $M > 1$, so case-03 is satisfied.

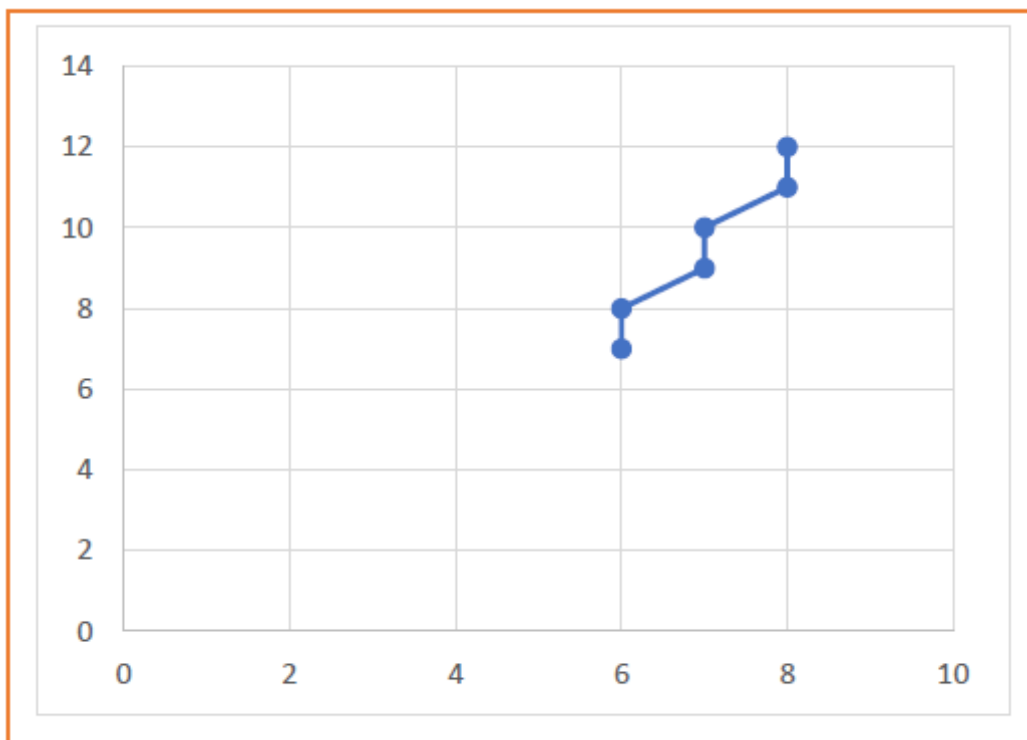
Now, Step-02 is executed until Step-03 is satisfied.

- $X_1 = X_0 + 1/M$

- $Y_1 = Y_0 + 1$

$$1/M = .5$$

X_p	Y_p	X_{p+1}	Y_{p+1}	Round off (X_{p+1}, Y_{p+1})
5	6	5.5	7	(6, 7)
		6	8	(6, 8)
		6.5	9	(7, 9)
		7	10	(7, 10)
		7.5	11	(8, 11)
		8	12	(8, 12)



Problem-02:

Calculate the points between the starting point (5, 6) and ending point (13, 10).

Solution-

Given-

- Starting coordinates = $(X_0, Y_0) = (5, 6)$
- Ending coordinates = $(X_n, Y_n) = (13, 10)$

Step-01:

Calculate ΔX , ΔY and M from the given input.

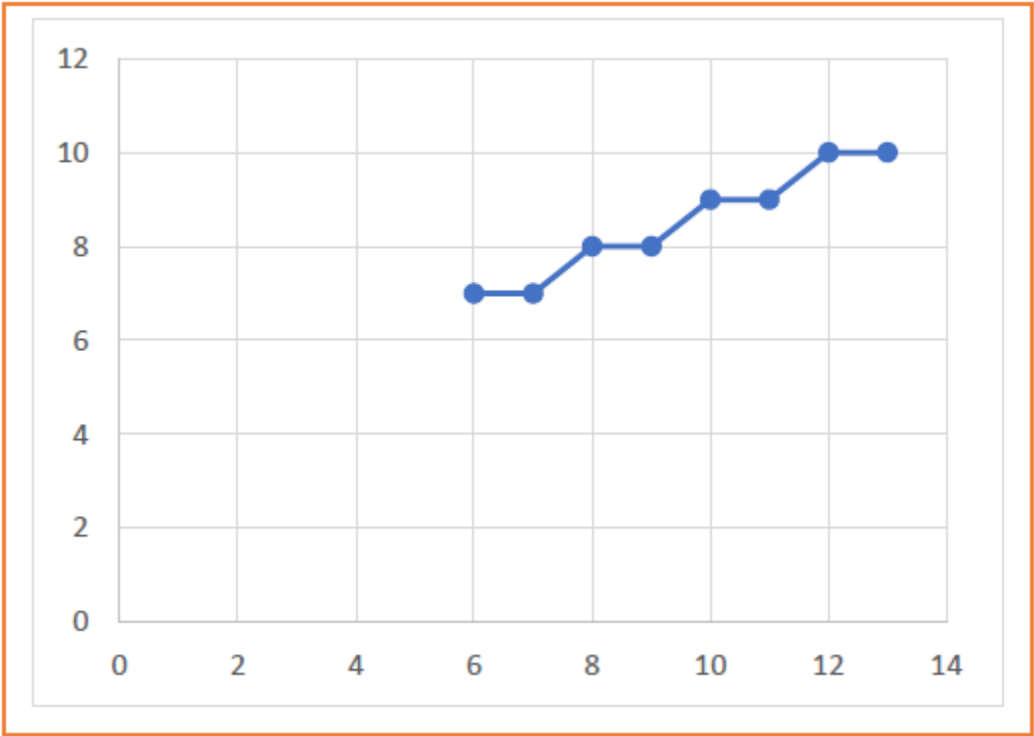
- $\Delta X = X_n - X_0 = 13 - 5 = 8$
- $\Delta Y = Y_n - Y_0 = 10 - 6 = 4$
- $M = \Delta Y / \Delta X = 4 / 8 = 0.50$

Step-02:

As $M < 1$, so case-01 is satisfied.

Now, Step-02 is executed until Step-03 is satisfied.

X_p	Y_p	X_{p+1}	Y_{p+1}	Round off (X_{p+1}, Y_{p+1})
5	6	6	6.5	(6, 7)
		7	7	(7, 7)
		8	7.5	(8, 8)
		9	8	(9, 8)
		10	8.5	(10, 9)
		11	9	(11, 9)
		12	9.5	(12, 10)
		13	10	(13, 10)



Problem-03:

Calculate the points between the starting point (1, 7) and ending point (11, 17).

Solution-

Given-

- Starting coordinates = $(X_0, Y_0) = (1, 7)$
- Ending coordinates = $(X_n, Y_n) = (11, 17)$

Step-01:

Calculate ΔX , ΔY and M from the given input.

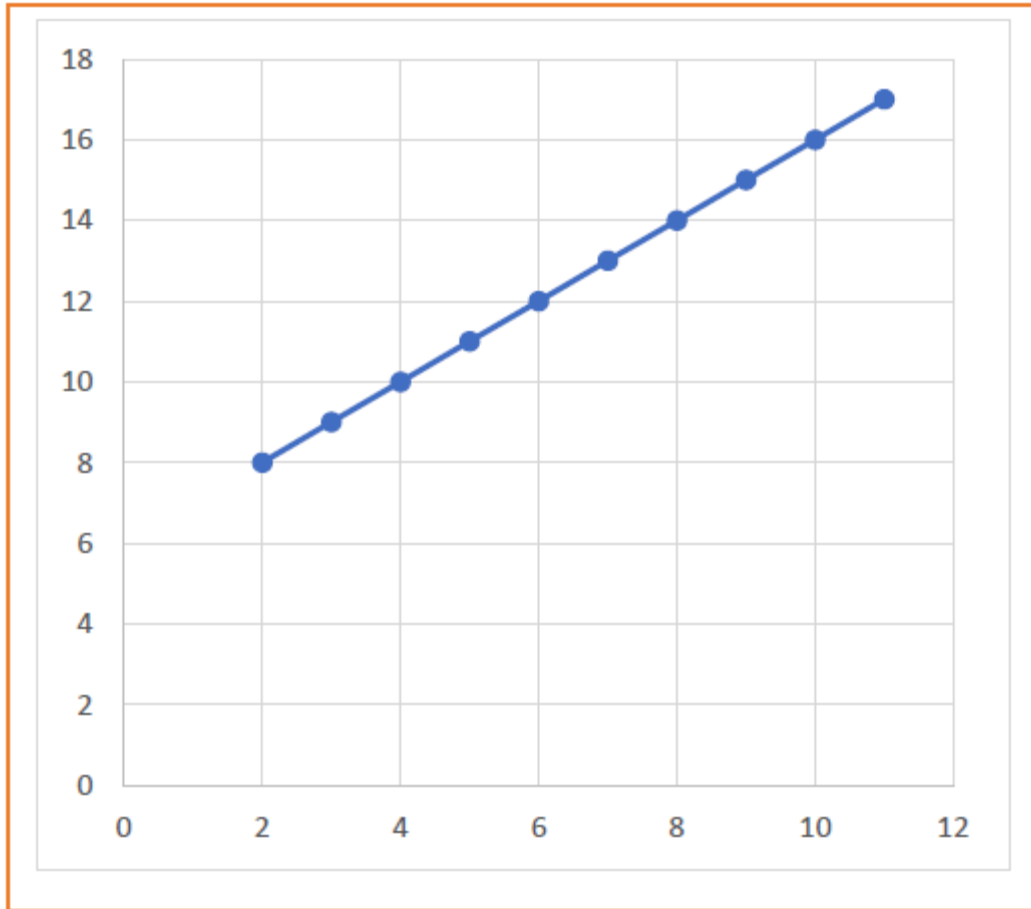
- $\Delta X = X_n - X_0 = 11 - 1 = 10$
- $\Delta Y = Y_n - Y_0 = 17 - 7 = 10$
- $M = \Delta Y / \Delta X = 10 / 10 = 1$

Step-02:

As $M = 1$, so case-02 is satisfied.

Now, Step-02 is executed until Step-03 is satisfied.

X_p	Y_p	X_{p+1}	Y_{p+1}	Round off (X_{p+1}, Y_{p+1})
1	7	2	8	(2, 8)
		3	9	(3, 9)
		4	10	(4, 10)
		5	11	(5, 11)
		6	12	(6, 12)
		7	13	(7, 13)
		8	14	(8, 14)
		9	15	(9, 15)
		10	16	(10, 16)
		11	17	(11, 17)



Advantages of DDA Algorithm-

The advantages of DDA Algorithm are-

- It is a simple algorithm.
- It is easy to implement.
- It avoids using the multiplication operation which is costly in terms of time complexity.

Disadvantages of DDA Algorithm-

The disadvantages of DDA Algorithm are-

- There is an extra overhead of using round off() function.
- Using round off() function increases time complexity of the algorithm.
- Resulted lines are not smooth because of round off() function.
- The points generated by this algorithm are not accurate.